

Explainable Survival Prediction from Gene Expression and Clinical Data

xyz^a , xyz^b , $xyz^{a,*}$, xyz^a , xyz^a , xyz^b , xyz^a

^a Department of Computer Science and Engineering, Amity University Kolkata, Kolkata, West Bengal, India

^b Department of Computational Biology, Indraprastha Institute of Information Technology Delhi, New Delhi, India

* Corresponding author.

ABSTRACT

Survival estimates in oncology are usually built from the size and spread of the tumour, yet patients placed in the same stage live for very different lengths of time. Measurements of gene activity explain part of that spread, although models built on them are rarely readable at the level of individual genes. A survival model that stays inspectable at every stage was therefore assembled from clinical records and gene activity for 1,620 patients ($N = 1,620$), drawn from four public cohorts covering brain, liver, pancreatic and skin cancer. Redundant genes were removed first, which reduced 500 candidates to 59, and those were compressed into 50 summary variables by principal component analysis (PCA). A Cox proportional hazards model was then fitted with an elastic-net penalty, reaching a concordance index of 0.736 on a held-out test set of 324 patients; the concordance index measures how often two patients are ordered correctly. Because a relative risk score cannot be read as a probability, isotonic regression was fitted on a separate split to convert scores into survival probabilities, giving Brier scores between 0.132 and 0.178 across the four horizons (12, 24, 36 and 60 months). Uncertainty for each patient was estimated from 5,000 simulated survival times, not from a single point prediction, yielding pessimistic, median and optimistic bounds (P_{10} , P_{50} , P_{90}). Reversing the compression by one matrix multiplication then recovered an exact risk weight for each of the 59 genes, so no approximate explanation method was required at any point.

KEYWORDS

Survival analysis; gene expression; Cox proportional hazards; elastic net; probability calibration; model explainability

1 Introduction

Treatment decisions in oncology rest on an estimate of how long a patient is likely to live, and that estimate is usually built from the tumour, node and metastasis (TNM) staging system, which groups patients by tumour size and by how far the disease has spread [1]. Staging describes anatomy. It says nothing about the molecular activity inside the tumour, so two patients placed in the same stage often survive for very different lengths of time. Part of that spread can be traced to differences in gene activity, which ribonucleic acid sequencing (RNA-Seq) measures directly from a tissue sample [2]. Several genes measured this way already carry prognostic weight: high activity of CXCL5 is associated with shorter survival in liver cancer [3], and CA9 marks tumour regions starved of oxygen [4]. Since The Cancer Genome Atlas (TCGA) began releasing matched clinical records and gene activity measurements across dozens of cancer types [5], a direct question has become answerable. Can staging information and gene activity be combined into one model that predicts survival better than staging alone, while remaining transparent enough for a clinician to inspect?

The standard tool for this task remains the Cox proportional hazards model [6], which works with a hazard (the instantaneous risk of dying at a given moment, for a patient who is still alive). Rather than assume a shape for how that baseline risk changes over time, it assumes only that each patient's hazard is the baseline hazard multiplied by a fixed factor, derived from a weighted sum of their measurements. Right censoring is handled naturally, since a patient still alive when the study closes contributes the information that they survived at least that long. One important limit comes with this flexibility: the number of measurements (P) must stay well below the number of patients (N). As the two approach each other the fitted weights are no longer uniquely determined [7], and gene activity data breaches the limit immediately, since a single experiment measures thousands of genes for a few hundred patients.

Three families of methods have been developed to work around that limit, of which the first adds a penalty term to the Cox model that pushes weights towards zero. The lasso penalty does so aggressively enough to set some weights exactly to zero, thereby selecting a subset of genes [7]. Elastic-net regularisation adds a second term to that behaviour [8, 9]. The second term shrinks correlated genes together instead of picking one representative from each group at random [10, 11]. The second family reduces the number of variables before fitting, most often through PCA. That

replaces many correlated measurements with a few summary variables capturing most of the variation [12, 13], and cross-validated penalised Cox models on gene activity data follow exactly this pattern [14]. The third family corrects the probabilities a model reports, and isotonic regression does so without assuming any shape for the mapping, while preserving the ordering of patients that the model already established [15, 16]. Neural network models take a different route altogether. DeepSurv [17], Cox-nnet [18] and Cox-PASNet [19] replace the weighted sum with a flexible function that captures effects a linear predictor cannot. They also require more patients than a single cancer cohort usually provides, and their internal weights no longer map onto named genes.

Four problems persist when these methods are combined. First, groups of genes acting in the same biological pathway are frequently left in the model together, and because their measurements are almost identical, the fitted weights become unstable across repeated fits [10]. Second, the Cox model reports a relative risk score, from which no clinician can read a one-year or five-year survival probability without a separate conversion step that many pipelines omit. Third, a single predicted number conveys nothing about the range of outcomes plausible for one patient, which is precisely what surgical and palliative care planning requires. Fourth, PCA solves the variable-count problem but introduces a readability problem in its place, because the fitted weights attach to summary variables, not to named genes. Post-hoc methods such as SHapley Additive exPlanations (SHAP) [20] and Local Interpretable Model-agnostic Explanations (LIME) [21] estimate gene-level importance by sampling, and therefore approximate the answer instead of recovering it.

All four problems are addressed here within a single model, through five contributions. A correlation filter fitted on the training data alone removes redundant genes, reducing 500 candidates to 59. A Cox model with an elastic-net penalty is then fitted using a smoothed objective, which keeps the optimiser stable where the penalty is otherwise not differentiable. Horizon-specific isotonic regression calibrators, fitted on a split held back for the purpose, convert risk scores into survival probabilities. A simulation draws 5,000 possible survival times per patient, yielding pessimistic, median and optimistic estimates rather than a single number. Finally, the compression step is reversed by one matrix multiplication, returning an exact risk weight for each of the 59 genes together with an exact additive breakdown per patient. The resulting data flow is traced in Figure 1.

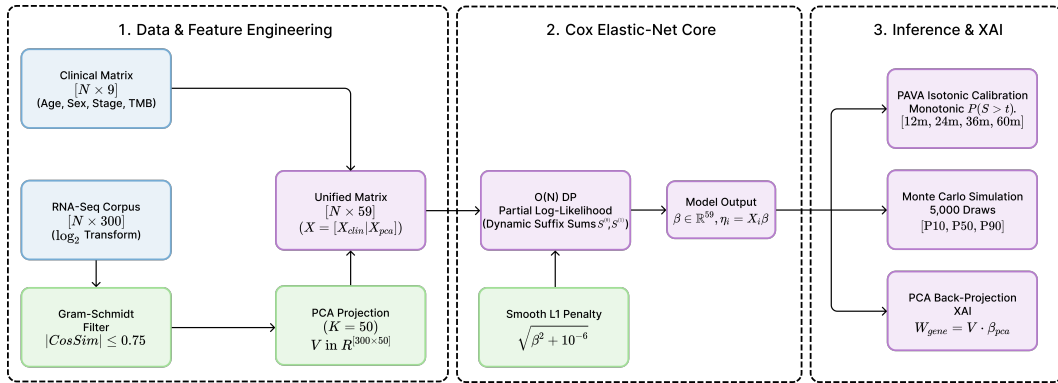


Figure 1: Overall data flow. Raw clinical records and gene activity measurements enter on the left, then pass through the correlation filter, the compression step, the penalised Cox model, the probability calibrator and the simulation stage, with gene-level explanations leaving on the right.

2 Literature Survey

2.1 Statistical models

Early work relating gene activity to survival extended the Cox model with penalty terms, applying ridge, lasso and elastic-net regularisation to the partial likelihood [7–10]. Because these methods retain the weighted-sum structure of the Cox model, each fitted weight continues to refer to one named measurement, and that property is what later deep models give up. Van Houwelingen et al. established the recipe still in use on microarray data: fit the penalty strength by cross-validation, then report performance on data the model never saw [14]. Ishwaran et al. took a different route with random survival forests [22], dropping the proportional hazards assumption altogether and estimating error from the samples left out of each tree.

2.2 Neural network models

Katzman et al. replaced the weighted sum in the Cox model with a neural network (DeepSurv) [17], and Ching et al. applied the same idea to gene activity data from ten TCGA cohorts (Cox-nnet) [18]. Hao et al. went further and imposed biological pathway structure on the network, evaluating Cox-PASNet on TCGA brain and ovarian cancer data [19]. Poirion et al. combined several deep and classical models into an ensemble across many TCGA cancer types (DeepProg) [23], which is notable less for the ensemble than for being tested on independent cohorts rather than by cross-validation alone. Dynamic-DeepHit extends the problem itself, using repeated measurements over time to update predictions as new data arrives [24]. Across this group the concordance index is almost always the headline metric, whereas calibrated probabilities, per-patient uncertainty and external validation are reported far less consistently.

2.3 Feature representation

Most models in this area combine gene activity data with clinical variables, and some add measurements of microRNA, deoxyribonucleic acid methylation or tissue images. Autoencoder models compress several such data types into a small set of learned variables. Chaudhary et al. applied this approach to 360 TCGA liver cancer patients ($n = 360$), reporting a concordance index of 0.68 with testing on independent liver cancer cohorts [25]. Learned variables of this kind capture patterns a weighted sum cannot. They are also difficult to trace back to individual genes, and that difficulty is the readability gap this work targets.

2.4 Comparison of existing models

The studies discussed above are summarised in Table 1, alongside the present work.

Table 1: Summary of related survival prediction studies.

Study	Cohort	Input data	Model	Validation and metric
van Houwelingen et al. [14]	Breast cancer, microarray	Gene expression	Penalised Cox	Cross-validation
Ishwaran et al. [22]	General survival data	Clinical and mixed	Random survival forest	Out-of-bag error
Katzman et al. [17]	Clinical cohorts	Clinical data	DeepSurv	Concordance index
Ching et al. [18]	Ten TCGA cohorts	Gene expression	Cox-nnet	Concordance index
Hao et al. [19]	TCGA brain and ovarian	Expression, clinical, pathway	Cox-PASNet	Concordance index
Chaudhary et al. [25]	TCGA liver, $n = 360$	Expression, microRNA, methylation	Autoencoder	Concordance index 0.68; independent cohorts
Poirion et al. [23]	32 TCGA cancers	Multiple data types	DeepProg	Concordance index; independent cohorts
Lee et al. [24]	Cystic fibrosis registry, $n = 5,883$	Repeated clinical measurements	Dynamic-DeepHit	Time-dependent discrimination
This work	Four TCGA cancers, $n = 1,620$	Expression and clinical	Cox elastic net	Held-out test split; concordance index, area under the curve, Brier score

2.5 Limitations of existing approaches

Several limitations recur across this literature. Sample sizes remain small relative to the number of measurements, which raises the risk of unstable weights in any method flexible enough to exploit them. Because TCGA records one tissue sample per patient at diagnosis, models requiring repeated measurements cannot be applied to it directly [24], and forcing a time-varying structure onto single-timepoint data means carrying values forward artificially, which introduces immortal time bias [26]. Learned variables from autoencoders are hard to map back to named genes [25], while pooling several cancer types into one model averages gene effects that may differ between those types. Most importantly, discrimination is reported far more often than calibration or per-patient uncertainty, and external validation remains inconsistent even where its value has been demonstrated [23].

2.6 Research gap

Penalised Cox regression [7, 10], variable compression [14], probability calibration [27], uncertainty estimation [24] and gene-level explanation [19] have each been studied separately. No published pipeline combines all five, however, while keeping training, calibration and test data strictly separated and returning an exact rather than an approximate gene-level explanation. That combination is the contribution of this work, using one measurement per patient taken at diagnosis. A version drawing on repeated measurements is left for future work.

3 Proposed Method

3.1 Cohort assembly

Patients were drawn from four TCGA cohorts: glioblastoma multiforme (study code GBM), a brain cancer [28]; liver hepatocellular carcinoma (LIHC) [29]; pancreatic adenocarcinoma (PAAD) [30]; and skin cutaneous melanoma (SKCM) [31]. These four codes are dataset identifiers used by TCGA, not standard clinical abbreviations, a distinction worth stating because LIHC in particular is easily read as a disease abbreviation. Clinical records and RNA-Seq gene activity values were downloaded in the file format published by cBioPortal [32, 33], then merged into a single table of 1,620 patients ($N = 1,620$).

Clinical fields present in only one cohort were removed, the melanoma skin type field being the clearest example. That field is missing for every patient in three of the four cohorts, and filling those gaps with a single value would create an artificial cluster of identical entries.

The cohort was split into training ($n = 972$), calibration ($n = 324$) and test ($n = 324$) subsets, stratified on whether a death was observed rather than on survival time. Stratifying on survival time under right censoring shifts the distribution between splits, whereas stratifying on the observed event held the death rate nearly constant, at 58.4% (training), 58.6% (calibration) and 58.3% (test).

3.2 Removing redundant genes

The 500 candidate gene activity columns were screened before any scaling or compression took place, with absolute Pearson correlations computed between every pair of those columns using the training split alone. The columns were then examined in a fixed order, and any column correlating above 0.75 with an earlier column was discarded, so that the rule keeps the first member of each correlated group and drops the rest.

On this cohort 36,205 gene pairs exceeded the threshold, and the filter discarded 441 of the 500 genes while keeping 59. Those discards were not arbitrary. The strongest pairs were genes already known to act together: the two acyl-coenzyme A synthetase genes (ACSM2A, ACSM2B) correlated at $r = 0.996$, and the three fibrinogen chain genes (FGA, FGB, FGG) correlated above $r = 0.98$ with one another. That pattern indicates the filter removed genuine biological redundancy, not random noise. One correlation matrix is the whole cost, which avoids the repeated matrix inversions a full variance inflation factor (VIF) audit would require. Correlation structure before and after this step is shown for a representative gene subset in Figure 2.

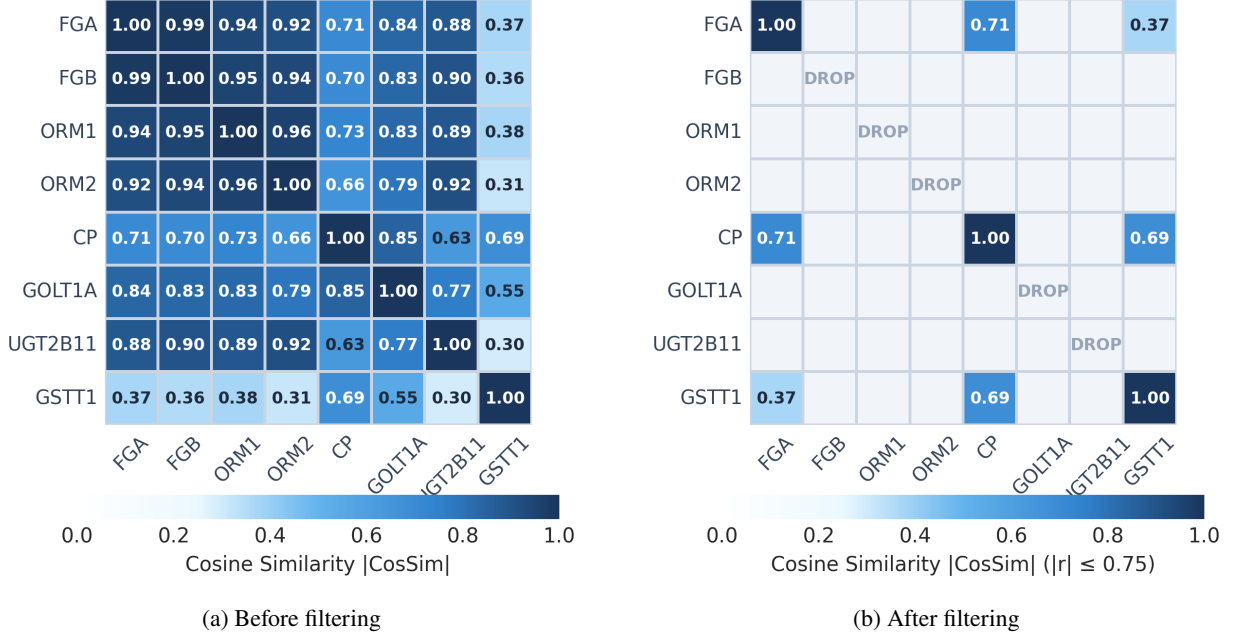


Figure 2: Pairwise correlation between a representative subset of genes, measured on the training split. Dark cells on the left mark groups of genes that move together, such as the fibrinogen chain genes; on the right the discarded genes are marked and only weakly correlated genes remain.

3.3 Scaling and compression

Because a small number of genes dominate any sequencing library, RNA-Seq counts follow a heavily right-skewed distribution [34], so a base-two logarithm with a pseudocount of one, $\log_2(x + 1)$, was applied to the 59 surviving columns. The pseudocount both shortens that tail and keeps the transform defined where activity is zero.

Transformed values were then scaled to zero mean and unit variance, producing the matrix $Z_{\text{gene}} \in \mathbb{R}^{N \times 59}$, with the scaling computed inside each cross-validation training fold only. Computing it on all data first would leak information from the validation fold into training.

Scaling has to precede compression. Because PCA finds the directions of largest variation [12, 13], an unscaled gene whose numbers simply span a wider range would dominate the first direction regardless of its biological role. The loading matrix $V \in \mathbb{R}^{59 \times 50}$ (satisfying $V^T V = I_{50}$) was computed by singular value decomposition, and the scaled genes were then projected onto $K = 50$ summary variables, giving $X_{\text{pca}} = Z_{\text{gene}} V \in \mathbb{R}^{N \times 50}$. How the surviving genes map onto the leading summary variables is shown in Figure 4.

3.4 Model input and objective

Clinical fields were processed on a separate branch, where sex, race, ethnicity, cancer type and age group were one-hot encoded (turning each category into its own binary column) and age was filled with the median where missing, then scaled. The result was a clinical block $X_{\text{clin}} \in \mathbb{R}^{N \times 20}$, joined to the 50 summary variables and rescaled to give the model input $X = [X_{\text{clin}} \mid X_{\text{pca}}] \in \mathbb{R}^{N \times 70}$.

The Cox model assigns each patient i a risk score $\eta_i = X_i \beta$ with hazard $h(t \mid X_i) = h_0(t) \exp(\eta_i)$ [6], the weights being $\beta = [\beta_{\text{clin}} \mid \beta_{\text{pca}}] \in \mathbb{R}^{70}$. Fitting requires the Breslow partial log-likelihood [35], which if evaluated directly needs a nested loop over patients at every observed death, at a cost of $O(N^2)$ operations. Death times were instead sorted in descending order and running totals of the exponentiated risk scores accumulated in a single pass, reducing the cost to $O(N)$. An elastic-net penalty was then added to the negative partial log-likelihood [8, 9]: its absolute-value term drives weights of uninformative variables to zero, while its squared term shrinks correlated variables together.

3.5 Optimisation

The objective was minimised with the limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm with bound constraints (L-BFGS-B) [36], which requires a gradient everywhere. The absolute-value term has no gradient at zero, and it produced undefined values during development. It was therefore replaced by the smooth approximation $\sqrt{\beta^2 + \epsilon}$

with $\epsilon = 10^{-6}$. That value is small enough to preserve the sparsifying behaviour near zero, yet far enough above double-precision rounding error to keep the gradient stable. The risk score η_i was additionally clipped to $[-40, 40]$ before exponentiation, since the upper bound already represents an extreme relative risk ($\exp(40) \approx 2.35 \times 10^{17}$) while remaining well below the double-precision overflow point ($\exp(709)$).

Two hyperparameters were tuned by grid search under five-fold nested cross-validation, with the penalty strength α searched over $\{0.1, 0.3, 0.8, 1.5, 3.0\}$ and the balance between the two penalty terms ($l1_ratio$) over $\{0.0, 0.1, 0.3, 0.5, 0.7\}$. The search selected $\alpha = 3.0$ and $l1_ratio = 0.7$, the final fit converged after 51 iterations, and preprocessing and cross-validation were implemented with scikit-learn [37]. How the fitted weights and loadings feed the later stages is shown in Figure 3.

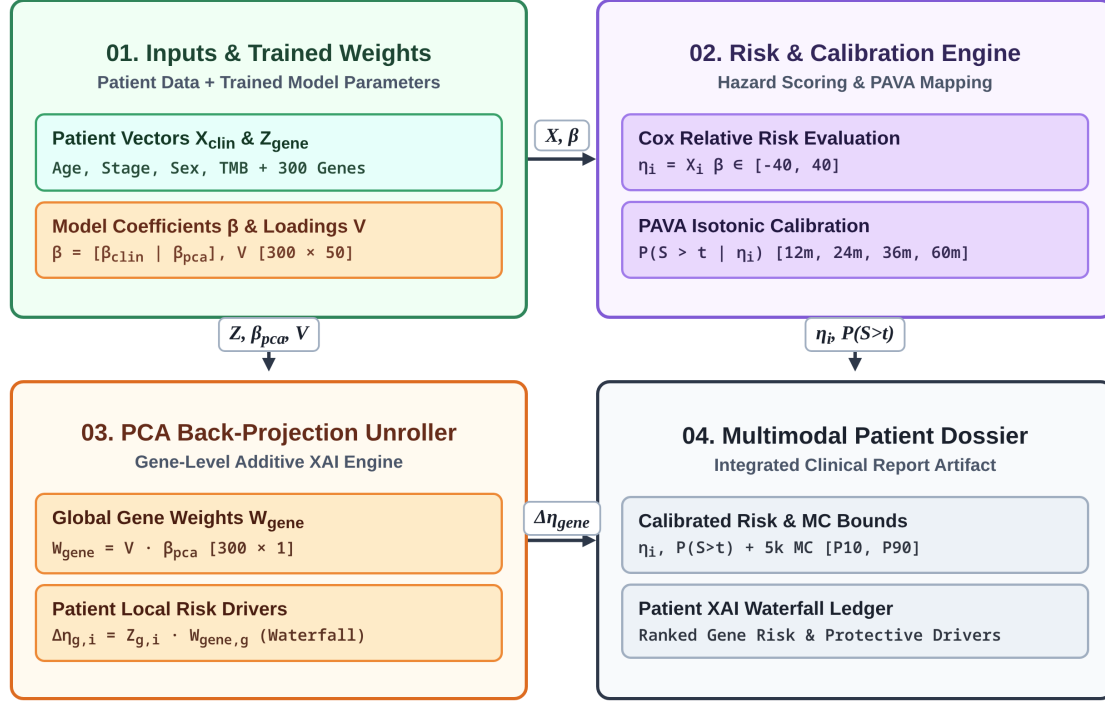


Figure 3: Flow through the prediction and explanation stages. Fitted weights and compression loadings feed the risk score, the probability calibrator and the reversal step that produces gene-level and patient-level explanations.

3.6 Converting risk scores into probabilities

The risk score η_i is meaningful only relative to other patients, so it does not answer the question a patient actually asks, which is the chance of surviving past a given date. The calibration split ($n = 324$) was held out before any filtering or scaling took place, and isotonic regression was fitted on it [15, 16, 27]. Because isotonic regression fits a step function that never decreases, it cannot reverse the ordering the Cox model already produced. Separate calibrators were fitted at 12, 24, 36 and 60 months, each mapping a risk score to a survival probability against the observed survival fractions from the Kaplan-Meier estimator [38]. The four fitted mappings are plotted in Figure 5.

3.7 Estimating uncertainty by simulation

A single predicted number conveys nothing about the spread of possible outcomes, so individual survival times were simulated against the Breslow cumulative baseline hazard $\Lambda_0(t)$ estimated from the training split [35]. For each patient, 5,000 independent values $U^{(k)} \sim \text{Uniform}(0, 1)$ were drawn and converted to a target hazard, $\Lambda_{\text{target}} = -\ln(U^{(k)})/\exp(\eta_i)$, with the matching month located by binary search over the baseline hazard step function (an $O(\log E)$ lookup, where $E = 461$ is the number of distinct observed death times).

The 5,000 draws per patient were summarised as a pessimistic bound ($P10$), a median ($P50$) and an optimistic bound ($P90$), and the simulated survival curve was integrated to 60 months to give the restricted mean survival time [39]. Integration was capped at the longest observed follow-up in the cohort (218.4 months), so that the simulation never extrapolates beyond the data. Example bands are shown in Figure 6.

3.8 Recovering gene-level weights exactly

PCA reduced 59 genes to 50 summary variables, so the fitted weights attach to those summary variables, not to individual genes. Both the compression and the Cox risk score are linear operations, however, so the compression can be reversed exactly. The sub-vector of weights attached to the summary variables is $\beta_{\text{pca}} \in \mathbb{R}^{50}$, and multiplying the loading matrix by this sub-vector gives a risk weight for every surviving gene:

$$W_{\text{gene}} = V \cdot \beta_{\text{pca}} \in \mathbb{R}^{59}. \quad (1)$$

For patient i and gene g , the contribution to the risk score follows directly from the scaled activity value $Z_{g,i}$ that entered the compression step:

$$\Delta\eta_{g,i} = Z_{g,i} \cdot W_{\text{gene},g}. \quad (2)$$

This is exact linear algebra, not an estimate, so the contributions of all 70 model inputs add up to the patient’s risk score with no residual. Sorting them by size produces a breakdown that can be checked term by term (Figure 7). Sampling-based methods such as SHAP [20] and LIME [21] estimate the same quantity, but only approximately.

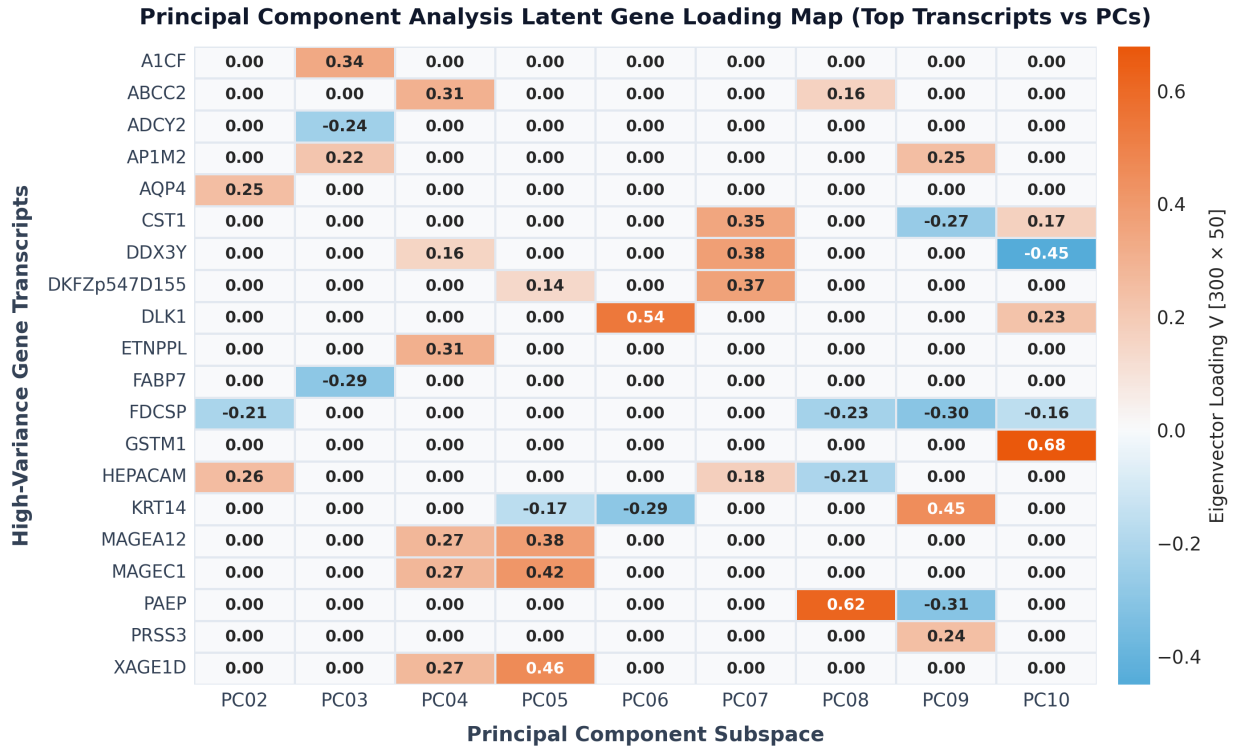


Figure 4: Loadings of a representative subset of the 59 surviving genes on the first ten summary variables. Cell values give the size of each gene’s contribution to the corresponding summary variable.

4 Results

4.1 Feature reduction

The model input shrank in stages, beginning with 506 candidate columns (500 gene activity columns and 6 clinical fields). The correlation filter left 65 columns, of which 59 were genes and 6 clinical. One-hot encoding then expanded the 6 clinical fields into 20 columns, and PCA compressed the 59 genes into 50 summary variables, giving a final model input of 70 columns (Table 2).

Removing 441 of 500 genes amounts to discarding most of the panel (88.2%). An unfiltered 500-gene panel therefore carries largely repeated information, not independent signal. That supports filtering before compressing, instead of relying on PCA alone to absorb the redundancy.

Table 2: Number of columns at each stage of the pipeline.

Stage	Columns
Candidate gene activity columns	500
Clinical fields, before encoding	6
Gene columns after correlation filter	59
Clinical fields after correlation filter	6
Summary variables after compression	50
Clinical columns after one-hot encoding	20
Final model input X	70

4.2 Model performance

Five-fold nested cross-validation on the training split selected $\alpha = 3.0$ and $l1_ratio = 0.7$, with a mean cross-validated concordance index of 0.722 ± 0.021 across folds. The concordance index measures how often the model ranks two patients in the correct survival order [40], where 0.5 matches random ordering and 1.0 perfect ordering; censoring-adjusted variants of the statistic have also been proposed [41].

The final model reached a concordance index of 0.749 on the training split and 0.746 on the calibration split, falling to 0.736 on the held-out test split ($n = 324$; Table 3). A gap of only 0.013 between training and test performance indicates limited overfitting, which is consistent with the aggressive filtering and shrinkage applied upstream.

Table 3: Concordance index by data split.

Split	Concordance index
Training ($n = 972$)	0.749
Calibration ($n = 324$)	0.746
Test ($n = 324$)	0.736
Five-fold nested cross-validation (mean $\pm$ s.d.)	0.722 ± 0.021

Inspection of the fitted weights shows which inputs mattered most. Cancer type carried the largest clinical effect, with membership of the brain cancer cohort giving the largest risk-increasing weight ($\beta = 0.772$) and membership of the melanoma cohort the largest protective weight ($\beta = -0.322$). That ordering matches the observed survival in these cohorts (Table 5). Patient age gave the third-largest clinical weight ($\beta = 0.293$, risk increasing). Among the compressed gene variables, the first summary variable gave the largest weight ($\beta = 0.304$).

The two blocks contributed comparable amounts. Summed across all 50 summary variables, the absolute weights of the gene block totalled 2.814, against 2.086 for the 20 clinical columns. Both blocks were scaled the same way, so these totals are directly comparable. By this measure the compressed gene signal carried slightly more weight than the full clinical panel. The ten largest weights across both blocks are ranked in Figure 7(a).

4.3 Calibration at fixed time horizons

The calibrators were evaluated on the test split ($n = 324$), restricted at each horizon to patients whose outcome at that horizon is known after accounting for right censoring. That left 289, 247, 228 and 218 patients at 12, 24, 36 and 60 months respectively (Table 4).

Area under the curve measures discrimination at a fixed time point [42], and it rose from 0.759 at 12 months to a peak of 0.860 at 24 months before settling near 0.847 at 36 and 60 months. The Brier score measures the squared difference between the predicted probability and the observed outcome, and is therefore better when lower [43]; it fell steadily from 0.178 at 12 months to 0.132 at 60 months.

The 60-month horizon carried both the lowest error and the highest observed death rate (78.4%), and that the death rate rises with horizon length has a mechanical cause. As the window extends, fewer patients remain who were censored before the horizon, so more of the uncertainty at 12 months comes from patients last seen alive and event-free. The fitted mapping at each horizon is plotted in Figure 5, where each curve rises without ever falling, confirming that the calibrator preserved the ordering the Cox model produced.

Table 4: Calibration results on the held-out test split.

Horizon (months)	Patients with known outcome	Death rate	Area under curve	Brier score
12	289	0.284	0.759	0.178
24	247	0.543	0.860	0.152
36	228	0.662	0.847	0.150
60	218	0.784	0.847	0.132

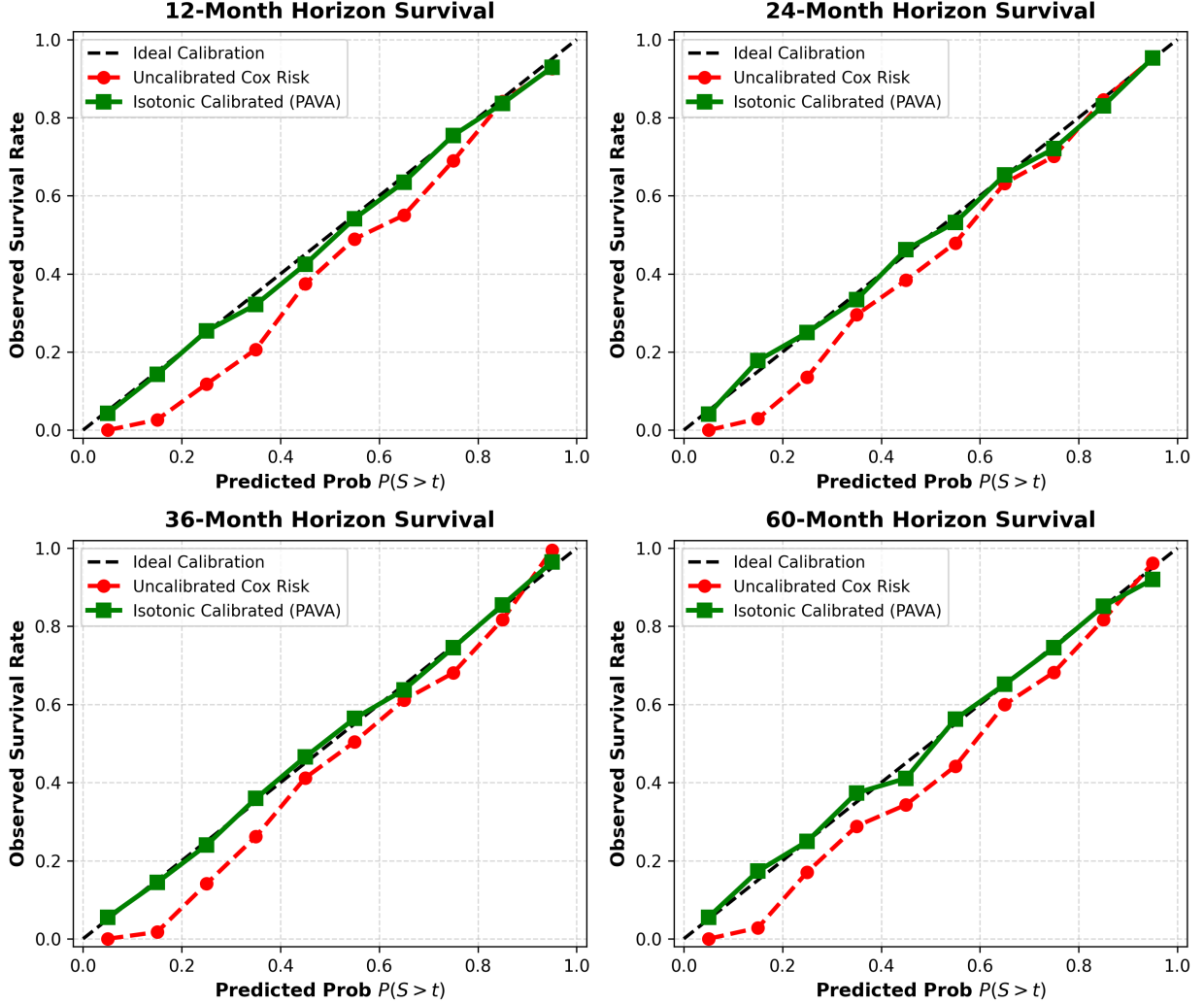


Figure 5: Fitted calibration curves at (a) 12 months, (b) 24 months, (c) 36 months and (d) 60 months, each comparing the uncalibrated Cox risk score with the calibrated survival probability on the test split. The diagonal marks perfect calibration.

4.4 Simulated uncertainty

The simulation produced individual P_{10} , P_{50} and P_{90} survival bounds for all 324 test patients, with a mean interval width between P_{10} and P_{90} of 107.8 months and a median width of 103.0 months. Intervals this wide reflect the difficulty of projecting a single tissue sample forward across a follow-up window reaching 218.4 months.

Interval width varied systematically with risk. One high-risk test patient ($\eta_i = +0.965$) received bounds of $P_{10} = 3.3$, $P_{50} = 13.6$ and $P_{90} = 38.1$ months, with a 60-month restricted mean survival time of 16.9 months. One low-risk patient ($\eta_i = -1.578$) received far wider bounds of $P_{10} = 20.8$ and $P_{50} = 152.2$ months. That patient's P_{90} reached the 218.4-month cap set by the longest observed follow-up, and the restricted mean survival time was 52.3 months.

The asymmetry follows from the exponential form of the hazard. A large positive risk score compresses the simulated death times towards the present, whatever value $U^{(k)}$ takes. A large negative score spreads the same distribution

across the whole follow-up range (Figure 6). In practice the P_{10} bound supplies a worst-case timeline for planning aggressive treatment, and the P_{90} bound a best-case ceiling limited by the cohort’s own follow-up (218.4 months), not by unbounded extrapolation.

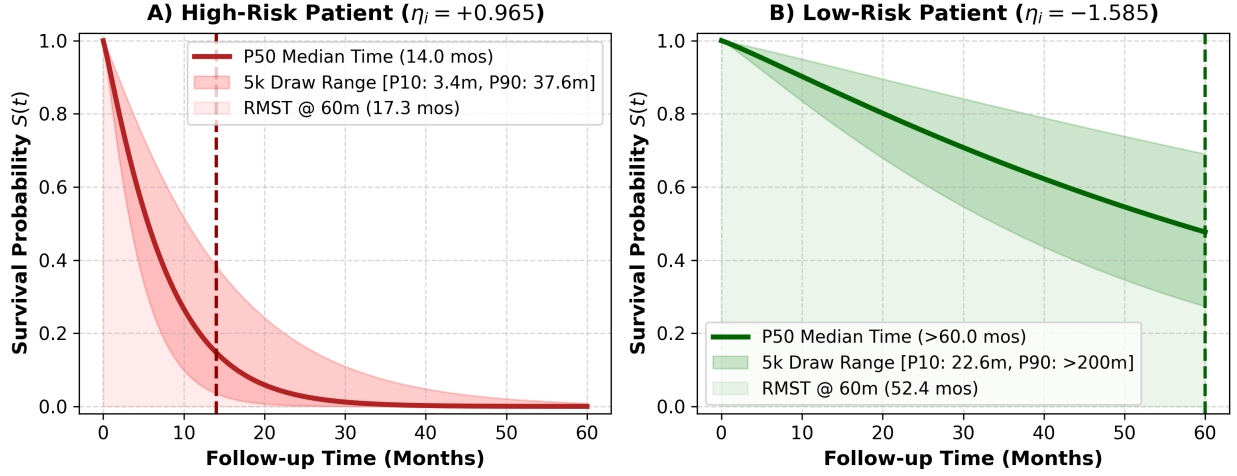


Figure 6: Simulated survival probability over a 60-month window for one high-risk and one low-risk test patient.

4.5 Gene-level and patient-level explanations

Reversing the compression resolved the first summary variable into its constituent genes, the ten loading most strongly being CD24, CXCL14, GPRC5A, PPP1R1B, SYT13, SERPINA3, CA9, ADAM6, FGFR2 and CXCL5. Two of these carry documented links to low-oxygen tumour regions and to spread beyond the original site: CA9 [4] and CXCL5 [3].

At the level of one patient, the breakdown splits the risk score across all 70 model inputs with no residual. A representative test patient (TCGA-DD-AACJ) is shown in Figure 7(b), still alive at last contact after 69.0 months of follow-up. The largest single term for this patient was protective, not belonging to the brain cancer cohort contributing -0.616 , while the two largest risk-increasing terms were patient age ($+0.328$) and one summary variable ($+0.321$). All 70 terms summed to the patient’s risk score of -0.028 , and because the decomposition is exact matrix algebra, each of those numbers can be audited individually.

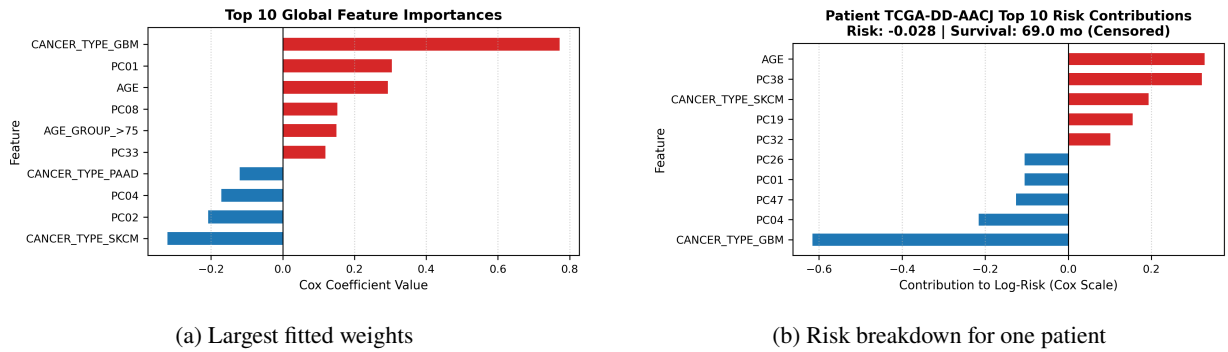


Figure 7: Explanations produced by reversing the compression step. (a) Inputs ranked by fitted weight across the whole cohort. (b) Additive breakdown of one patient’s risk score.

4.6 Independent statistical screen

The raw clinical and gene activity columns were also tested for association with mortality independently of the model. Two tests were applied: a two-sided Mann-Whitney U test against whether a death was observed [44], and a Spearman rank correlation against survival time in months [45]. Both were corrected for multiple testing across all 1,049 evaluated columns of the 1,620-patient cohort by the Benjamini-Hochberg procedure [46]. Since both tests examine one variable at a time, they show association rather than an effect independent of the other variables.

Cancer type was strongly associated with mortality ($p = 1.09 \times 10^{-53}$, corrected $q = 1.36 \times 10^{-51}$, rank-biserial effect size 0.392), which matches the median survival across the pooled cohort, ranging from 12.16 months for brain cancer to 35.78 months for melanoma (Table 5). Tumour stage recorded under the American Joint Committee on Cancer system was also strongly associated ($p = 8.74 \times 10^{-45}$, corrected $q = 5.15 \times 10^{-43}$), so stage retains predictive value even inside a pooled analysis of four cancer types.

Among individual gene activity columns, DKFZp547D155 showed the strongest correlation with survival time ($\rho = -0.346$, corrected $q = 7.57 \times 10^{-31}$). Several genes recovered by reversing the compression also passed this screen independently, with CA9 at $\rho = -0.338$ (corrected $q = 7.23 \times 10^{-30}$) and CXCL5 at $\rho = -0.308$ (corrected $q = 4.33 \times 10^{-25}$). Because the screen makes no use of the fitted model, that agreement provides a check on the model-derived gene ranking.

Table 5: Outcome by cancer type across the pooled cohort ($N = 1,620$).

Cancer type	TCGA code	n	Death rate	Median survival (months)
Glioblastoma multiforme	GBM	595	0.827	12.16
Pancreatic adenocarcinoma	PAAD	184	0.543	15.34
Liver hepatocellular carcinoma	LIHC	372	0.355	19.76
Skin cutaneous melanoma	SKCM	469	0.475	35.78

5 Discussion

5.1 Why a linear model was chosen

A neural network could process all 500 candidate genes without a correlation filter or a compression step, but on a cohort of this size that flexibility is expensive. With 972 training patients, a network large enough to capture interactions between genes carries more trainable parameters than it has training examples. The usual defence is heavy regularisation, which in practice can push the network back towards a linear function.

The model reported here keeps a linear structure from the outset. Redundant genes were removed before they could destabilise the fitted weights, and the smoothed elastic-net penalty then shrank the remaining correlated variables together rather than selecting one representative from each group [8]. The resulting test concordance index of 0.736 sits close to the 0.68 reported by Chaudhary et al. for an autoencoder on 360 TCGA liver cancer patients [25]. The two numbers come from different cohorts under different splits, however, so this is a reference point rather than a controlled comparison.

Reversing the compression extends readability past the summary variables and back onto named genes, and that computation is exact where sampling-based explainers only estimate [20, 21]. An exact decomposition matters wherever the same explanation must be reproducible across repeated runs.

5.2 Clinical readability

A relative risk score can tell a clinician that one patient is at higher risk than another, but not what that patient’s survival probability is over a stated period. The calibration layer supplies that conversion, with Brier scores between 0.132 and 0.178 across the four horizons, and the simulation layer then converts a probability into a range of plausible survival months.

The asymmetry of those ranges is itself informative, being tight for high-risk patients and wide for low-risk ones, which suggests the model is confident about near-term decline in aggressive cases yet appropriately uncertain about long-horizon outcomes in more favourable ones.

Simulation was preferred to a confidence interval on the fitted weights, because a confidence interval on β describes a population-level parameter whereas the simulation estimates a distribution for one patient’s survival time. Since the baseline hazard step function ends at the cohort’s longest observed follow-up (218.4 months), the simulation cannot project beyond the available data.

5.3 Limitations

Six limitations should be stated. First, the proportional hazards assumption was not tested: Schoenfeld residuals provide the standard check [47] and were not computed, and the assumption is likely violated for at least some of the four cancer types given their different progression timelines.

Second, both selected hyperparameters sat at the upper edge of their search grids. $\alpha = 3.0$ was the largest penalty strength searched, and $l1_ratio = 0.7$ the largest balance value. A stronger penalty or a more lasso-weighted balance may therefore perform better, and was never explored.

Third, pooling four histologically different cancers into one weight vector averages gene effects across them, so a gene protective in one cancer type and harmful in another will have those effects cancel, trading subtype-specific precision for coverage across cancer types.

Fourth, the model has not been tested outside TCGA, and any use beyond this retrospective setting would require testing against batch effects and different sequencing platforms [23]. Fifth, each patient is represented by one tissue sample taken at diagnosis, so disease progression over time is not modelled, and methods that use repeated measurements need data TCGA does not provide [24]. Sixth, the statistical screen tested one variable at a time, and therefore shows association with mortality without establishing that any variable adds predictive value beyond the others.

6 Conclusion

A Cox proportional hazards model with an elastic-net penalty was fitted to clinical records and gene activity measurements from four pooled TCGA cancer cohorts, having first removed redundant genes to reduce 500 candidates to 59 and compressed those into 50 summary variables. The resulting 70-column model reached a concordance index of 0.736 on a held-out test split of 324 patients, and isotonic regression converted the raw risk scores into survival probabilities with Brier scores as low as 0.132. A 5,000-draw simulation against the Breslow baseline hazard then turned those probabilities into per-patient ranges of plausible survival months. Reversing the compression by matrix multiplication returned an exact risk weight for each of the 59 genes, so no approximate explainer was required. A separate single-variable screen across all 1,049 columns recovered several of the same genes, including CA9 and CXCL5. Three steps follow from this work: interaction terms specific to each cancer type should be added, the pipeline should be tested on cohorts outside TCGA, and the proportional hazards assumption should be checked with Schoenfeld residuals before any clinical use is considered.

Acknowledgement

[Insert acknowledgement here.]

References

- [1] Mahul B. Amin, Stephen B. Edge, Frederick L. Greene, David R. Byrd, Robert K. Brookland, Mary Kay Washington, Jeffrey E. Gershenwald, Carolyn C. Compton, Kenneth R. Hess, Daniel C. Sullivan, J. Milburn Jessup, James D. Brierley, Laurie E. Gaspar, Richard L. Schilsky, Charles M. Balch, David P. Winchester, Elliot A. Asare, Martin Madera, Donna M. Gress, and Laura R. Meyer, editors. *AJCC Cancer Staging Manual*. Springer, Cham, 8 edition, 2017. doi: 10.1007/978-3-319-40618-3.
- [2] Bo Li and Colin N. Dewey. Rsem: Accurate transcript quantification from rna-seq data with or without a reference genome. *BMC Bioinformatics*, 12:323, 2011. doi: 10.1186/1471-2105-12-323.
- [3] Su-Lin Zhou, Zhi Dai, Zheng-Jun Zhou, Xin-Yu Wang, Guo-Huan Yang, Zheng Wang, Xiao-Wu Huang, Jia Fan, and Jian Zhou. Overexpression of cxcl5 mediates neutrophil infiltration and indicates poor prognosis for hepatocellular carcinoma. *Hepatology*, 56(6):2242–2254, 2012. doi: 10.1002/hep.25907.
- [4] Alexandra Giatromanolaki, Michael I. Koukourakis, Efthimios Sivridis, Jaromir Pastorek, Charles C. Wykoff, Kevin C. Gatter, and Adrian L. Harris. Expression of hypoxia-inducible carbonic anhydrase-9 relates to angiogenic pathways and independently to poor outcome in non-small cell lung cancer. *Cancer Research*, 61(21): 7992–7998, 2001. URL <https://aacrjournals.org/cancerres/article/61/21/7992/508977>.

- [5] John N. Weinstein, Eric A. Collisson, Gordon B. Mills, Kenna R. Mills Shaw, Brad A. Ozenberger, Kyle Ellrott, Ilya Shmulevich, Chris Sander, and Joshua M. Stuart. The cancer genome atlas pan-cancer analysis project. *Nature Genetics*, 45(10):1113–1120, 2013. doi: 10.1038/ng.2764.
- [6] David R. Cox. Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2):187–202, 1972. doi: 10.1111/j.2517-6161.1972.tb00899.x.
- [7] Robert Tibshirani. The lasso method for variable selection in the cox model. *Statistics in Medicine*, 16(4): 385–395, 1997. doi: 10.1002/(SICI)1097-0258(19970228)16:4<385::AID-SIM380>3.0.CO;2-3.
- [8] Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005. doi: 10.1111/j.1467-9868.2005.00503.x.
- [9] Tong Tong Wu. Elastic net for cox’s proportional hazards model with a solution path algorithm. *Statistica Sinica*, 22:27–44, 2012. doi: 10.5705/ss.2010.107.
- [10] Noah Simon, Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Regularization paths for cox’s proportional hazards model via coordinate descent. *Journal of Statistical Software*, 39(5):1–13, 2011. doi: 10.18637/jss.v039.i05.
- [11] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2 edition, 2009. doi: 10.1007/978-0-387-84858-7.
- [12] Harold Hotelling. Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6):417–441, 1933. doi: 10.1037/h0071325.
- [13] Ian T. Jolliffe and Jorge Cadima. Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A*, 374(2065):20150202, 2016. doi: 10.1098/rsta.2015.0202.
- [14] Hans C. van Houwelingen, Tako Bruinsma, Augustinus A. M. Hart, Laura J. van’t Veer, and Lodewyk F. A. Wessels. Cross-validated cox regression on microarray gene expression data. *Statistics in Medicine*, 25(18): 3201–3216, 2006. doi: 10.1002/sim.2353.
- [15] Miriam Ayer, H. D. Brunk, G. M. Ewing, W. T. Reid, and Edward Silverman. An empirical distribution function for sampling with incomplete information. *The Annals of Mathematical Statistics*, 26(4):641–647, 1955. doi: 10.1214/aoms/1177728423.
- [16] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002. doi: 10.1145/775047.775151.
- [17] Jared L. Katzman, Uri Shaham, Alexander Cloninger, Jonathan Bates, Tingting Jiang, and Yuval Kluger. DeepSurv: Personalized treatment recommender system using a cox proportional hazards deep neural network. *BMC Medical Research Methodology*, 18:24, 2018. doi: 10.1186/s12874-018-0482-1.
- [18] Travers Ching, Xun Zhu, and Lana X. Garmire. Cox-nnet: An artificial neural network method for prognosis prediction of high-throughput omics data. *PLoS Computational Biology*, 14(4):e1006076, 2018. doi: 10.1371/journal.pcbi.1006076.
- [19] Jie Hao, Youngsoon Kim, Tejaswini Mallavarapu, Jung Hun Oh, and Mingon Kang. Interpretable deep neural network for cancer survival analysis by integrating genomic and clinical data. *BMC Medical Genomics*, 12(Suppl 10):189, 2019. doi: 10.1186/s12920-019-0624-2.
- [20] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, pages 4765–4774, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html.
- [21] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “why should i trust you?”: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016. doi: 10.1145/2939672.2939778.
- [22] Hemant Ishwaran, Udaya B. Kogalur, Eugene H. Blackstone, and Michael S. Lauer. Random survival forests. *The Annals of Applied Statistics*, 2(3):841–860, 2008. doi: 10.1214/08-AOAS169.

- [23] Olivier B. Poirion, Zheng Jing, Kumardeep Chaudhary, Sijia Huang, and Lana X. Garmire. Deepprog: An ensemble of deep-learning and machine-learning models for prognosis prediction using multi-omics data. *Genome Medicine*, 13:112, 2021. doi: 10.1186/s13073-021-00930-x.
- [24] Changhee Lee, Jinsung Yoon, and Mihaela van der Schaar. Dynamic-deephit: A deep learning approach for dynamic survival analysis with competing risks based on longitudinal data. *IEEE Transactions on Biomedical Engineering*, 67(1):122–133, 2020. doi: 10.1109/TBME.2019.2909027.
- [25] Kumardeep Chaudhary, Olivier B. Poirion, Liangqun Lu, and Lana X. Garmire. Deep learning-based multi-omics integration robustly predicts survival in liver cancer. *Clinical Cancer Research*, 24(6):1248–1259, 2018. doi: 10.1158/1078-0432.CCR-17-0853.
- [26] Samy Suissa. Immortal time bias in pharmacoepidemiology. *American Journal of Epidemiology*, 167(4):492–499, 2008. doi: 10.1093/aje/kwm324.
- [27] Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 625–632, 2005. doi: 10.1145/1102351.1102430.
- [28] Cameron W. Brennan, Roel G. W. Verhaak, Aaron McKenna, Benito Campos, Houtan Noushmehr, Sofie R. Salama, Siyuan Zheng, Debyani Chakravarty, J. Zachary Sanborn, Samuel H. Berman, Rameen Beroukhi, Brady Bernard, Chang-Jiun Wu, Giannicola Genovese, Ilya Shmulevich, Jill Barnholtz-Sloan, Lihua Zou, Rahulsimham Vegesna, Sachet A. Shukla, Giovanni Ciriello, W. K. Yung, Wei Zhang, Carrie Sougnez, Tom Mikkelsen, Kenneth Aldape, Darell D. Bigner, Erwin G. Van Meir, Michael Prados, Andrew Sloan, Keith L. Black, Jennifer Eschbacher, Gaetano Finocchiaro, William Friedman, David W. Andrews, Abhijit Guha, Mary Iacocca, Brian P. O’Neill, Greg Foltz, Jerome Myers, Daniel J. Weisenberger, Robert Penny, Raju Kucherlapati, Charles M. Perou, D. Neil Hayes, Richard Gibbs, Marco Marra, Gordon B. Mills, Eric Lander, Paul Spellman, Richard Wilson, Chris Sander, John Weinstein, Matthew Meyerson, Stacey Gabriel, Peter W. Laird, David Haussler, Gad Getz, and Lynda Chin. The somatic genomic landscape of glioblastoma. *Cell*, 155(2):462–477, 2013. doi: 10.1016/j.cell.2013.09.034.
- [29] The Cancer Genome Atlas Research Network. Comprehensive and integrative genomic characterization of hepatocellular carcinoma. *Cell*, 169(7):1327–1341, 2017. doi: 10.1016/j.cell.2017.05.046.
- [30] The Cancer Genome Atlas Research Network. Integrated genomic characterization of pancreatic ductal adenocarcinoma. *Cancer Cell*, 32(2):185–203, 2017. doi: 10.1016/j.ccell.2017.07.007.
- [31] The Cancer Genome Atlas Network. Genomic classification of cutaneous melanoma. *Cell*, 161(7):1681–1696, 2015. doi: 10.1016/j.cell.2015.05.044.
- [32] Ethan Cerami, Jianjiong Gao, Ugur Dogrusoz, Benjamin E. Gross, Selcuk Onur Sumer, Bülent Arman Aksoy, Anders Jacobsen, Caitlin J. Byrne, Michael L. Heuer, Erik Larsson, Yevgeniy Antipin, Boris Reva, Arthur P. Goldberg, Chris Sander, and Nikolaus Schultz. The cbio cancer genomics portal: An open platform for exploring multidimensional cancer genomics data. *Cancer Discovery*, 2(5):401–404, 2012. doi: 10.1158/2159-8290.CD-12-0095.
- [33] Jianjiong Gao, Bülent Arman Aksoy, Ugur Dogrusoz, Gideon Dresdner, Benjamin Gross, S. Onur Sumer, Yichao Sun, Anders Jacobsen, Rileen Sinha, Erik Larsson, Ethan Cerami, Chris Sander, and Nikolaus Schultz. Integrative analysis of complex cancer genomics and clinical profiles using the cbioportal. *Science Signaling*, 6(269):p11, 2013. doi: 10.1126/scisignal.2004088.
- [34] Michael I. Love, Wolfgang Huber, and Simon Anders. Moderated estimation of fold change and dispersion for rna-seq data with deseq2. *Genome Biology*, 15(12):550, 2014. doi: 10.1186/s13059-014-0550-8.
- [35] Norman E. Breslow. Discussion of the paper by d. r. cox. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2):216–217, 1972. doi: 10.1111/j.2517-6161.1972.tb00900.x.
- [36] Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyu Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995. doi: 10.1137/0916069.
- [37] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL <https://jmlr.org/papers/v12/pedregosa11a.html>.

- [38] Edward L. Kaplan and Paul Meier. Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282):457–481, 1958. doi: 10.1080/01621459.1958.10501452.
- [39] Patrick Royston and Mahesh K. B. Parmar. Restricted mean survival time: An alternative to the hazard ratio for the design and analysis of randomized trials with a time-to-event outcome. *BMC Medical Research Methodology*, 13:152, 2013. doi: 10.1186/1471-2288-13-152.
- [40] Frank E. Harrell, Robert M. Califf, David B. Pryor, Kerry L. Lee, and Robert A. Rosati. Evaluating the yield of medical tests. *JAMA*, 247(18):2543–2546, 1982. doi: 10.1001/jama.1982.03320430047030.
- [41] Hajime Uno, Tianxi Cai, Michael J. Pencina, Ralph B. D’Agostino, and L. J. Wei. On the c-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. *Statistics in Medicine*, 30(10):1105–1117, 2011. doi: 10.1002/sim.4154.
- [42] Patrick J. Heagerty and Yingye Zheng. Survival model predictive accuracy and roc curves. *Biometrics*, 61(1):92–105, 2005. doi: 10.1111/j.0006-341X.2005.030814.x.
- [43] Erika Graf, Claudia Schmoor, Willi Sauerbrei, and Martin Schumacher. Assessment and comparison of prognostic classification schemes for survival data. *Statistics in Medicine*, 18(17-18):2529–2545, 1999. doi: 10.1002/(SICI)1097-0258(19990915/30)18:17/18<2529::AID-SIM274>3.0.CO;2-5.
- [44] Henry B. Mann and Donald R. Whitney. On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1):50–60, 1947. doi: 10.1214/aoms/1177730491.
- [45] Charles Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904. doi: 10.2307/1412159.
- [46] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- [47] David Schoenfeld. Partial residuals for the proportional hazards regression model. *Biometrika*, 69(1):239–241, 1982. doi: 10.1093/biomet/69.1.239.